Gonia · Ionic volute · saved compass experiment

The compass changes centre.

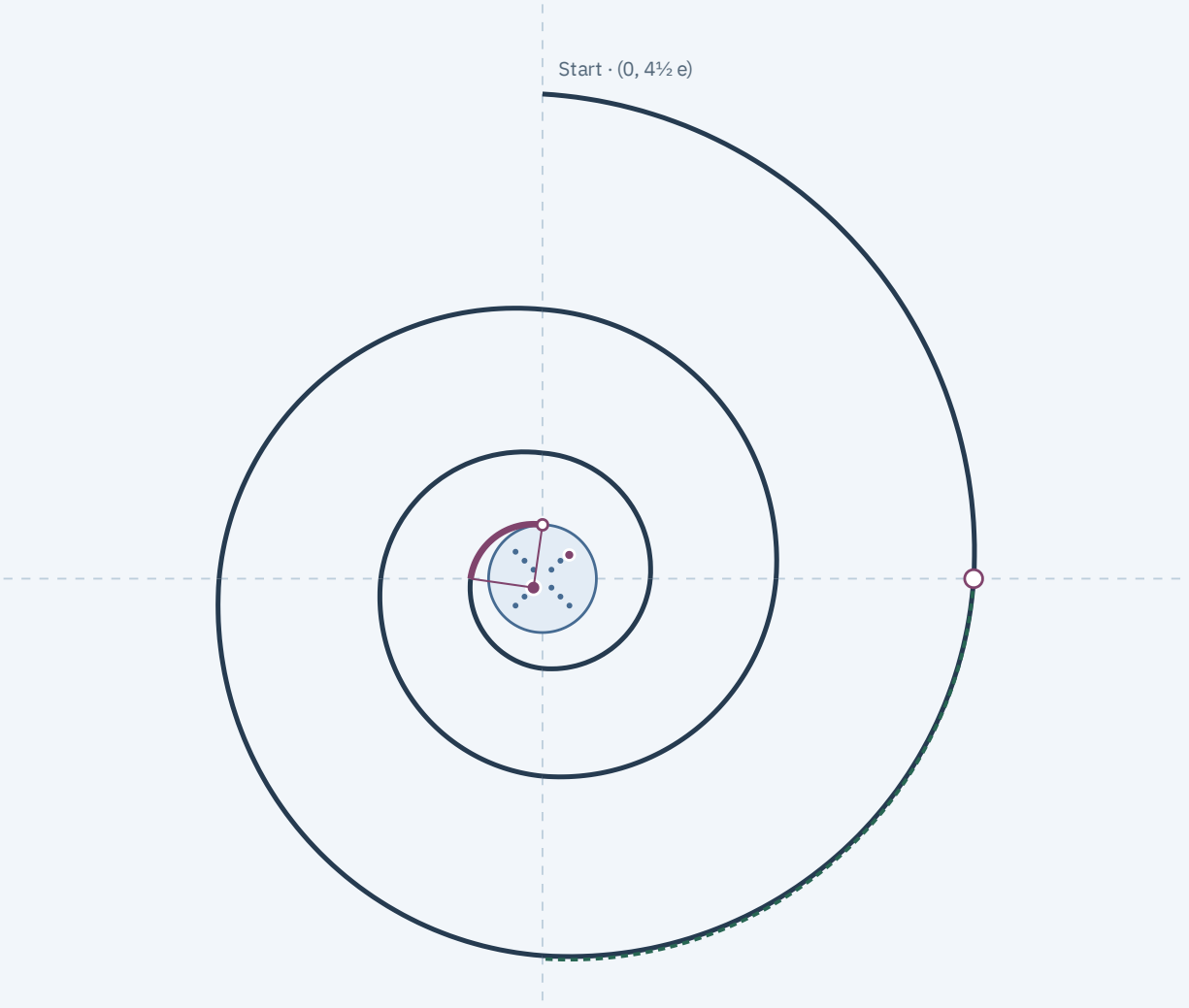
A modern exact quarter-turn interpretation of Juglaris & Locke’s 1889 English Vignola, plate XV. The twelve centres and named axis points are interpreted as ideal geometry. This is neither a survey of an ancient capital nor a recovered drawing by Vitruvius. The printed line’s precision cannot be inferred from our calculation.

Your drawing sequence

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 10 → 11 → 12. These choices follow the printed centre order so far. All committed arcs turn clockwise through 90°. Endpoint (0, 0.5) e. The endpoint lies exactly on the eye circle.

12/12 quarter circles drawn

Equal length scales · e = eye diameter



1889 printed guide → ideal circular arcs. New drawing, not an ancient capital survey.

Dashed plum is an uncommitted next-arc preview. A green overlay, when selected, changes only one next centre for inspection; it does not replace the committed outline.

Inspect a direction

Directions at actual angles

Incoming tangent rotated to horizontal; both rotate together.



`0.447614° at this join`

Solid plum: arriving. Dashed blue: departing. Green: aligned.

The displayed angle is calculated between two radii at the endpoint. Rotating both radii through 90° gives their tangent directions without changing the angle. Two arcs meeting at a point need not have the same tangent.

Committed quarter circles. Coordinates are exact twelfths of e; square roots and angles are rounded here. The CSV retains the integer squared radii and cross/dot products.

Arc / pin	Centre (e)	Radius (e)	End (e)	Next direction
1 / 1	-0.25, 0.25	4.257347	4, 0	0.447614°
2 / 2	0.25, 0.25	3.758324	0, -3.5	0.584631°
3 / 3	0.25, -0.25	3.259601	-3, 0	0.795724°
4 / 4	-0.25, -0.25	2.76134	0, 2.5	1.108812°
5 / 5	-0.166667, 0.166667	2.339278	2.166667, 0	0.678025°
6 / 6	0.166667, 0.166667	2.006932	0, -1.833333	0.946951°
7 / 7	0.166667, -0.166667	1.674979	-1.5, 0	1.414423°
8 / 8	-0.166667, -0.166667	1.34371	0, 1.166667	2.726311°
9 / 9	-0.083333, 0.083333	1.086534	1, 0	0.795724°
10 / 10	0.083333, 0.083333	0.920447	0, -0.833333	1.145763°
11 / 11	0.083333, -0.083333	0.754615	-0.666667, 0	1.789911°

Arc / pin	Centre (e)	Radius (e)	End (e)	Next direction
12 / 12	-0.083333, -0.083333	0.589256	0, 0.5	8.130102° to eye circle*

* An eye-circle direction comparison is a join only when the endpoint lies on the eye. Your final radial gap is 0 e. The local repair overlay does not alter this table or the committed arcs.

The ancient words and the later plate

Selected Latin: Krohn 1912, pp. 71–72. Line wraps and end-of-line hyphens are removed; punctuation is lightly regularized. Angle-bracketed editorial supplies remain. New English translations and grammar notes are learning aids. The 1889 English guide is a separate later interpretation.

Eight parts belong to the volutes

tunc ex novem partibus et dimidia una pars et dimidia abaci crassitudo relinquatur; reliquae octo volutis constituantur.

Then, from the nine and a half parts, let one and a half parts remain for the abacus's thickness; let the other eight be assigned to the volutes.

relinquatur and constituantur are passive commands: 'let … be left', 'let … be assigned'. The singular pars and plural octo [partes] take different verb endings. The total is 1½ + 8 = 9½, not nine parts.

Put the eye between two unequal lengths

tunc in eo loco, qui locus dividit quattuor et dimidiam et tres et dimidiam partem, centrum oculi ⟨conlocetur⟩

Then, at the place dividing four and a half from three and a half parts, let the centre of the eye be placed.

oculi is genitive: 'of the eye'. The eye is not halfway down the eight-part height. Krohn supplies ⟨conlocetur⟩, 'let it be placed'; the brackets remain visible here. This is an editorial supply, not a newly discovered ancient word.

Give the eye one part

signeturque ex eo centro rotunda circinatio tam magna in diametro, quam una pars ex octo partibus est. ea erit oculi magnitudine

And from that centre let a circular outline be marked, as large in diameter as one part out of the eight parts. That will be the size of the eye.

ex eo centro means ‘from that centre’. diametro names the diameter, not the radius. On this desk e is that diameter, so the eye’s radius is e/2. Changing the print size changes millimetres, not these ratios.

The reduction has a textual problem

tunc ab summo sub abaco inceptum ⟨schema volutae⟩ in singulis tetrantorum actionibus dimidiatum oculi spatium minuatur, doneque in eundem tetrantem, qui subest, conveniat.

Then let the ⟨volute’s figure⟩, begun from the top under the abacus, be diminished by half the eye’s span in each operation of the quadrants, until it comes into the same quadrant which lies below.

This deliberately close English follows Krohn’s difficult wording, not Morgan’s different ‘under the abacus’ ending. Krohn supplies ⟨schema volutae⟩ and edits qui subest, conveniat; his apparatus records alternatives. A phrase about quadrants does not itself locate twelve compass centres. Do not extend the half-eye decrement through three turns and call it the whole surviving recipe.

A promised figure

de volutarum descriptionibus, uti ad circinum sint recte involutae, quemadmodum describantur, in extremo libro forma et ratio earum erit subscripta.

Concerning the drawing of the volutes—how they are to be drawn so as to be wound correctly to the compass—their figure and calculation will be set down at the end of the book.

forma et ratio means ‘figure and calculation/account’. The future erit points forward. The ancient illustration is not preserved here. The 1889 plate below is a much later construction, not the fulfilment of this promise in a surviving ancient manuscript.

Words used at the desk

voluta · volutarum
scroll · of the scrolls
A form related to winding or rolling. The title names an architectural scroll; our arc construction follows one selected later guide, not every Ionic volute.

oculus · oculi
eye · of the eye
The small circle at the centre. It supplies the unit e. It is not the same point as any of the twelve compass pins.

circinus · circini
compass · of the compass
A drawing compass has one fixed leg at the selected centre and a moving leg on the curve. Changing the fixed leg changes the circle.

minuatur
let it be diminished
Present passive subjunctive from minuere. This command gives a reduction; it does not name the geometric location of the next centre.

ἄβαξ · abacus
board or slab; the capital’s top member
Greek headword and its Latin architectural borrowing. This is vocabulary, not a Greek original of the Latin passage. The abacus lies above the volutes.

κάθετος · cathetus

perpendicular; a dropped vertical
Greek headword and the Latinized technical term. The vertical line through the eye locates its upper and lower portions. A construction pin need not lie on that line.

διάμετρος · diametros
diameter
Greek headword and the borrowing in Vitruvius. The diameter runs across a circle through its centre; confusing it with a radius doubles or halves the construction.

What the exact construction assumes

In units of e/12, the eye centre is (0,0), its radius is 6, and the start is (0,54). Pins 1–4 are (–3,3), (3,3), (3,–3), (–3,–3); pins 5–8 use ±2, and pins 9–12 use ±1. A quarter turn sends P to C + (P_y – C_y, –P_x + C_x). Integer coordinates remain exact. The radius is the square root of the retained integer squared radius. Join angles use atan2(|cross|, dot).

The full 1–12 sequence ends at (0,6), on the eye. This establishes contact, not tangent continuity. The selected local intervention projects the next centre onto the incoming radius; that makes one pair of directions agree and moves the next endpoint. It is a modern experiment, not a proposed historical restoration. The eye circle itself is not moved.

Reference: Maria Losito's 2012 research notice on Goldmann's different 1649 volute documents the distinction between later constructions. Our desk operates the specified 1889 plate, not Goldmann's construction.

Six complete pages and a surviving capital

Complete PDF pages; Poppler pdftoppm, longest side 2200 px, JPEG quality 88; no cropping or reconstruction. Public-domain editions; each included image and font was checked against its retained SHA-256 identity before this reading was produced.

uti, quam crassus imus scapus fuerit addita octava decuma parte scapi, abacus habeat longitudinem et latitudinem; crassitudinem cum volutis eius dimidiam. recedendum autem est ab extremo abaco in interiorem partem frontibus volutarum parte duodevicensima et eius dimidia. tunc crassitudo dividenda est in partes novem et dimidiam, et secundum abacum in quattuor partibus volutarum secundum extremi abaci quadram lineae demittendae, quae cathetoe dicuntur. tunc ex novem partibus et dimidia una pars et dimidia abaci crassitudo relinquatur, reliquae octo volutis constituantur. tunc ab linea, quae secundum abaci 6
extremam partem demissa erit, in interiorem partem ⟨alia⟩
79 recedat unius et dimidiatae partis latitudine. deinde hae lineae dividantur ita, ut quattuor partes et dimidia sub abaco relinquantur. tunc in eo loco, qui locus dividit quattuor et dimidiam et tres et dimidiam partem, centrum oculi ⟨conlocetur⟩ signeturque ex eo centro rotunda circinatio tam magna in diametro, quam una pars ex octo partibus est. ea erit oculi magnitudine, et in ea catheto respondens diametros agatur. tunc ab summo sub abaco inceptum ⟨schema volutae⟩ in singulis tetrantorum actionibus dimidiatum oculi spatium minuatur, doneque in eundem tetrantem, qui subest, conveniat. capituli autem 7
crassitudo sic est facienda, ut ex novem partibus et dimidia tres partes praependeant infra astragalum summi scapi; cymatio, adempto abaco et canali, reliqua sit pars. proiectura autem cymatii habeat extra abaci quadram oculi magnitudinem. pulvinorum baltei ab abaco hanc habe-

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1 octava decuma *HS*: nona decima *G* 2 parte *G*: partem *HS* 8 dimittendae *x* 12 demissa G^c: dimissa *x* *add. Joc* 15 relinquantur *G*: -quatur *HS* 17 conlocetur *add. Kr* (signetur ducaturque *Joc*) 19 catheto *Joc*: catheton *S*, cathecton *HG* 21 *add. Kr* (*cf. 66, 4; 136, 1 etc.*) 22 doneque (donique *Lachm*): denique *x* 23 qui subest, conveniat *Kr*: qui est sub
a
abconveniat (*H*, subaco veniat *G*, sub abaco veniat *S*) *x*
26 adempto *ed*: adepto *x* 27 habet *H et a. c. S* 28 ab *G*: *om. HS*

Vitruvius III.5.5–7 · Krohn 1912, p. 71. Eye proportions, quarter-turn reduction, and Krohn's critical apparatus. Angle brackets mark editorial supplies. PDF page 89. Original source

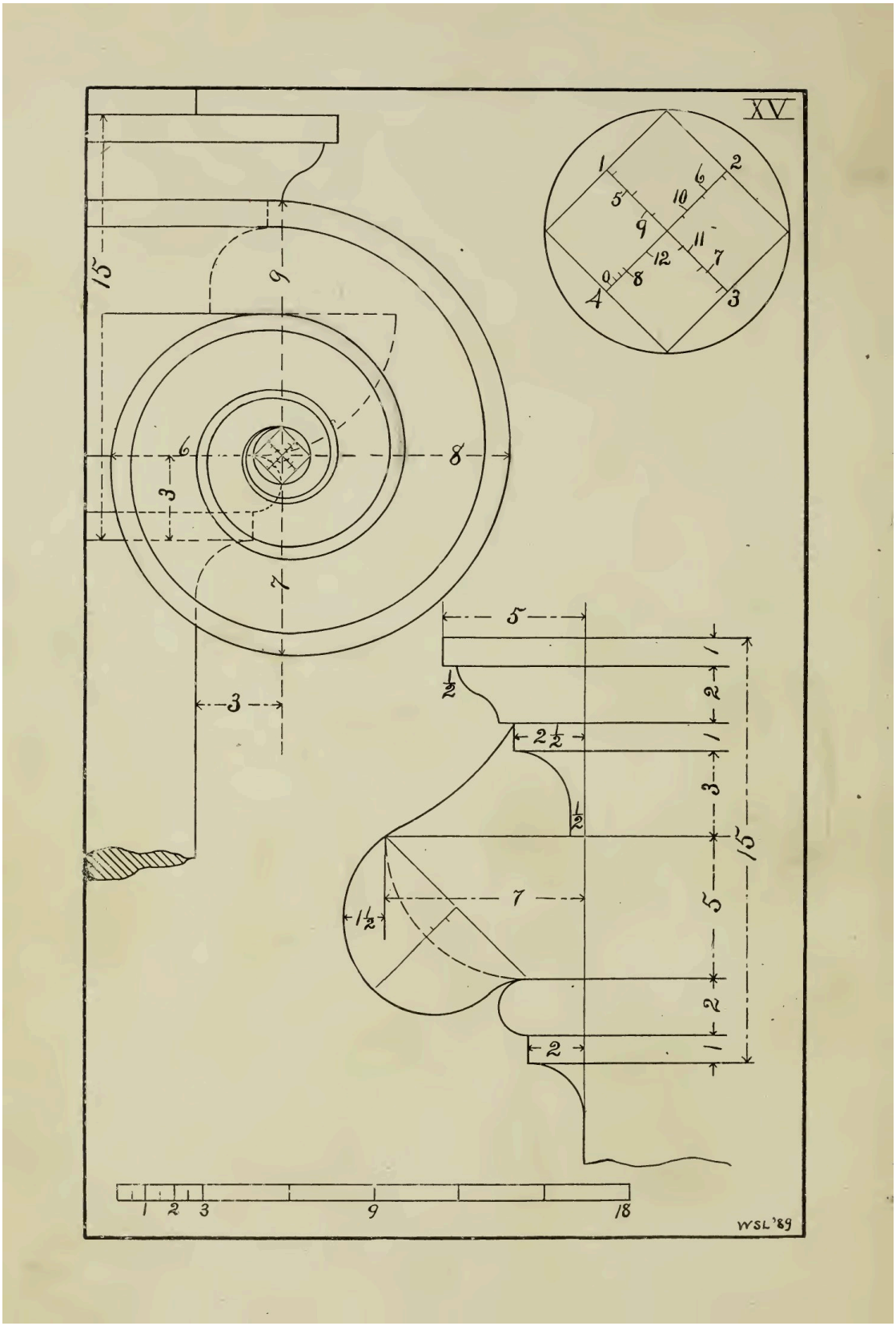
ant proiecturam, uti circini centrum unum cum sit positum in capituli tetrante et alterum diducatur ad extremum cymatium, circumactum balteorum extremas partes tangat. axes volutarum ne crassiores sint quam oculi magnitudo, volutaeque ipsae succidantur altitudinis suae duodecimam partem. (20) haec erunt symmetriae capitulorum, quae columnae futurae sunt ab minimo ad pedes XXV. quae supra erunt, reliqua habebunt ad eundem modum symmetrias, abacus autem erit longus et latus, quam crassa columna est ima adiecta parte VIIII, uti, quo minus habuerit altior columna contractum, 80 eo ne minus habeat capitulum suae symmetriae proiecturam 8 et in latitudine suae partis adiectionem. de volutarum descriptionibus, uti ad circinum sint recte involutae, quemadmodum describantur, in extremo libro forma et ratio earum (5) erit subscripta.

Capitulis perfectis, deinde ⟨in scapis⟩ columnarum non ad libellam sed ad aequalem modulum conlocatis, ut, quae adiectio in stylobatis facta fuerit, in superioribus membris respondeat [symmetria epistyliorum], epistyliorum ratio sic est habenda, uti, si columnae fuerint a minima XII pedum (10) ad quindecim pedes, epistylii sit altitudo dimidia crassitudinis imae columnae; item ⟨si⟩ ab XV pedibus ad XX, columnae altitudo dimetiatur in partes tredecim, et unius partis altitudo epistylii fiat; item si a XX ad XXV pedes, dividatur altitudo in partes XII et semissem, et eius una pars (15) epistylium in altitudine fiat; item si ab XXV pedibus ad XXX, dividatur in partes XII, et eius una pars altitudo fiat. item ⟨si quae altiores erunt,⟩ rata parte ad eundem modum ex altitudine columnarum expediendae sunt altitudines epi- 9 styliorum. quo altius enim scandit oculi species, non facile

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2 diducatur *G*: deducatur *HS* 4 ne *Joc*: nec *x* 5 succidantur *Mar*: sic cedantur *x* 6 capitulorumque *HS* 12 latitudine *rec*: altitudine *x* suae partis H^cSG^c: repartis *HG* 16 *add. Joc* 19 *del. Meister* 22 *add. Phil* quindecī *S* 23 demetiatur *x* 24 viginti *S* 25 *et* 27 duodecī *S* 28 *add. Ro* (*sic*: si quae supra altiores fuerint: *cf. 74, 22*) rata parte *Schn* (pro rata parte *Mar*): ratā partē (-tem *H*) *x*

Vitruvius III.5.7–8 · Krohn 1912, p. 72. The promise of a figure and calculation at the end of the book. No surviving ancient volute plate is supplied here. PDF page 90. Original source



Juglaris & Locke 1889 · plate XV. The later construction operated on this desk. The plate bears WSL '89. Its enlarged eye labels twelve compass centres. PDF page 68. Original source

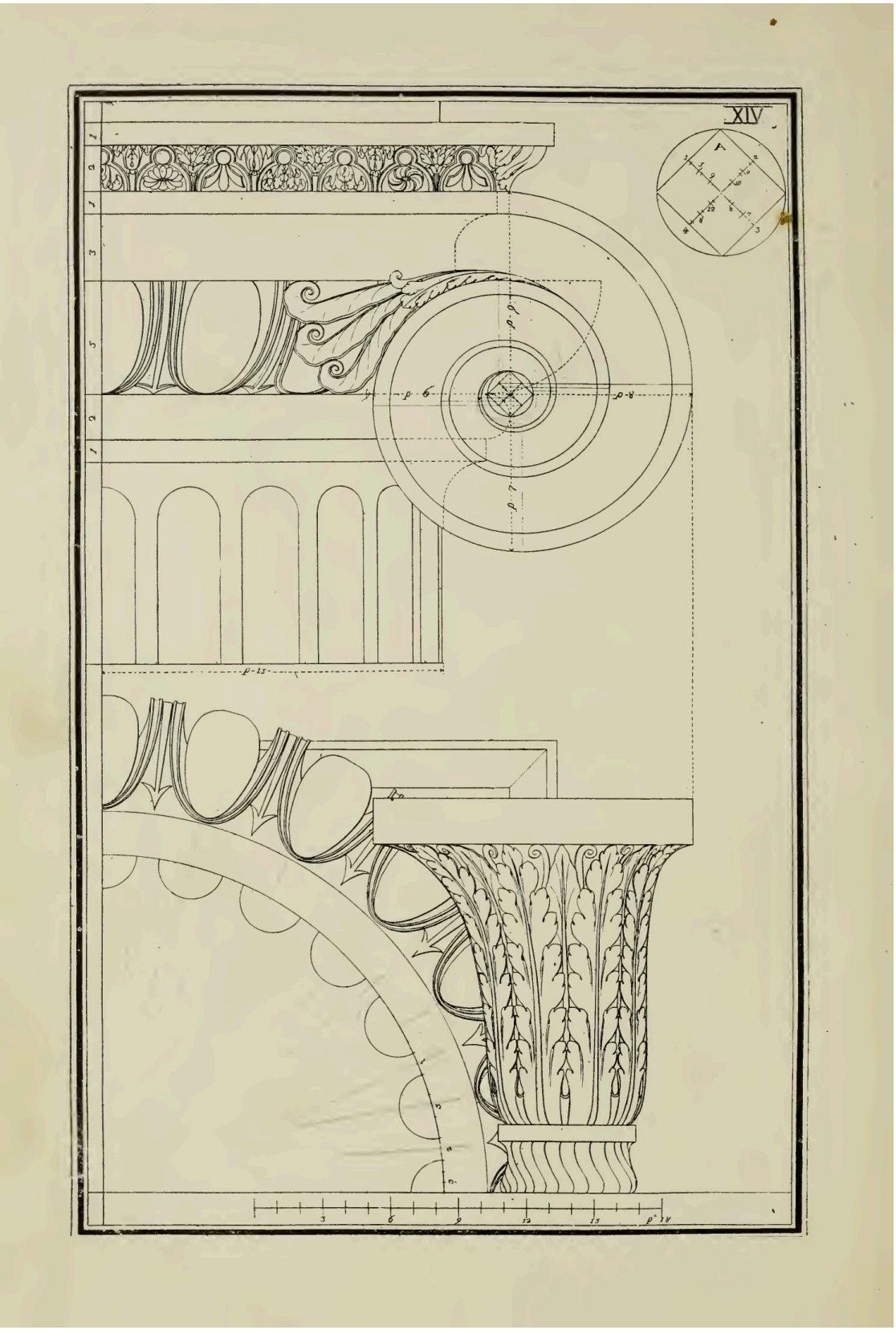
PLATE XV.

THE IONIC VOLUTE.

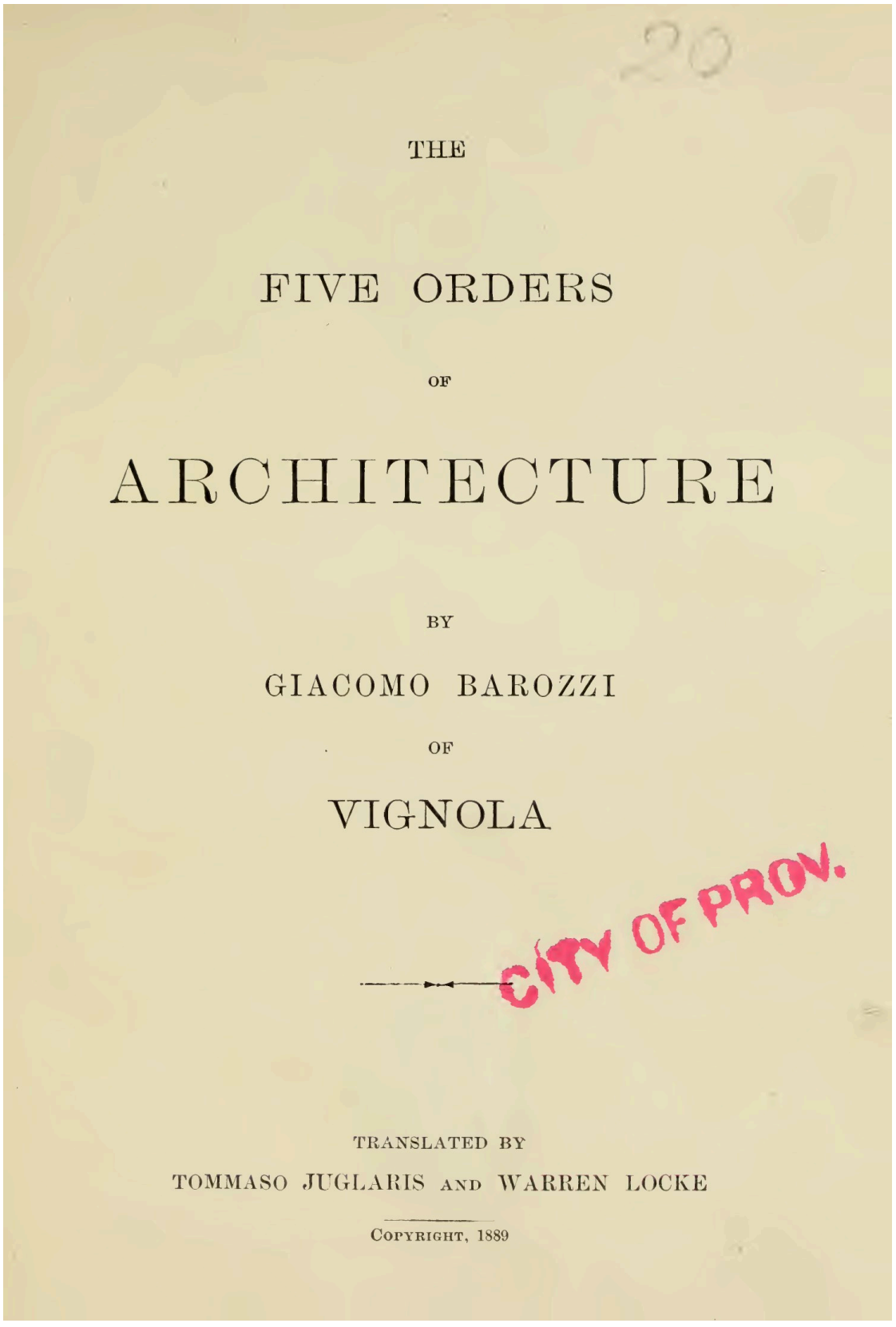
A DRAWING for the volute of an Ionic capital may be made as follows: Having the outline of the capital, the center of the volute is on the horizontal line of the top of the bead at the top of the column, and 3 parts from the side, and 18 parts from the center of the column. Draw the eye — a circle — 2 parts in diameter. Inscribe a square in this circle with perpendicular and horizontal diagonals. Bisect the sides with 45° lines through the center. The ends of these last are the four centers for the first turn of the volute. From the center of the eye lay off 9 parts above (just touching the ogee), 8 parts to the right, 7 parts below, and 6 parts to the left. From point 1 (see enlarged diagram of eye), with radius reaching the 9-point, strike an arc. It will pass through point 8. With radius 2, 8 make the next quarter, with 3, 7 make the third, and with 4, 6 complete the first turn of the volute. The second turn is made from centers 5, 6, 7, 8, which are $\frac{1}{3}$ of the half-diagonal from points 1, 2, 3, 4, and the third turn is made from points 9, 10, 11, 12. The inner line of the fillet of the volute is made to taper toward the outside line by using a center $\frac{1}{12}$ of a half-diagonal nearer the center of the eye (see O on diagram) than is used for the corresponding outside line.

The outlines and dimensions of the capital are given in the second drawing, which also shows the form of the roll of the volute midway between the two faces.

Juglaris & Locke 1889 · explanation of plate XV. The English instructions name the 9, 8, 7, 6 axis points, quarters and nested centre divisions. They are not ancient Latin. PDF page 69. Original source



Juglaris & Locke 1889 · plate XIV. The Ionic capital in context. This is the later book's drawing, not a photograph or survey of a Greek capital. PDF page 64. Original source



Juglaris & Locke 1889 · title page. The title page identifies the translators and 1889 edition. The downloaded PDF's name must not turn it into a 1562 scan. PDF page 5. Original source



A surviving capital · Sardis, The Met 26.59.1. Museum photograph, not a construction plate. Greek, ca. 300 BCE. The displayed column omits most of its shaft; parts of the shaft are restored and the profiled base below the torus is a copy. The capital's original position is uncertain. Our 1889 outline is not fitted to this object. The Met Open Access, CC0. Complete photograph retained byte for byte. Original source

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New code, diagrams, English and notes:

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